

You can use the VMM to create a new virtual machine, and install a GNU/Linux distribution using a .iso image. You can specify RAM, disk size and follow the installation steps for your particular distro. You can also import an existing .qcow2 disk image to use it as a virtual machine.

Ansible with libvirt-VM

The version of Ansible used for this article is given below:

```
$ ansible --version
ansible 2.2.1.0
  config file = /etc/ansible/ansible.cfg
  configured module search path = Default w/o overrides
```

If you have the `sshd` daemon running on your local machine, you can use Ansible to test it. For example, a ping test on the localhost is shown below:

```
$ ansible localhost -m ping
localhost | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
```

You can also check how long the system has been up and running using the following commands:

```
$ ansible localhost -a uptime
localhost | SUCCESS | rc=0 >>

11:00:20 up 4:09, 0 users, load average: 0.18, 0.14, 0.11
```

You can execute a shell command on the remote machine (localhost, in this case) as illustrated below:

```
$ ansible localhost -a "date"
localhost | SUCCESS | rc=0 >>
Sun Feb 5 11:24:53 IST 2017
```

The 'setup' command provides details of the remote target machine. A snippet output is provided below:

```
$ ansible localhost -m setup
localhost | SUCCESS => {
  "ansible_facts": {
    "ansible_all_ipv4_addresses": [
      "192.168.10.1",
      "192.168.5.6"
    ],
    "ansible_all_ipv6_addresses": [
      "fe90::fc24:ff:feb9:cb61",
      "ff80::5846:fac1:6afc:2e30"
    ],
  },
}
```

```
"ansible_architecture": "x86_64",
"ansible_bios_date": "06/12/2016",
"ansible_bios_version": "R00ET45W (1.20 )",
"ansible_cmdline": {
  "BOOT_IMAGE": "/vmlinuz-linux-libre",
  "cryptdevice": "/dev/sda1:cryptroot",
  "quiet": true,
  "root": "/dev/mapper/cryptroot",
  "rw": true
},
....
```

An Ubuntu 15.04 instance with VMM is used in the following examples with Ansible. The IP address of the instance is added to `/etc/hosts`:

```
192.168.122.250 ubuntu
```

The `/etc/ansible/hosts` file contains the following:

```
ubuntu
```

You can now do a ping test from the host to the Ubuntu VM using the following command sequence for the user 'xetex':

```
$ ansible ubuntu -m ping -u xetex --ask-pass
SSH password:
ubuntu | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
```

To avoid prompting for the password, you can add the localhost public SSH key to the VM, as follows:

```
$ ssh-copy-id -i ~/.ssh/id_rsa.pub xetex@ubuntu

/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed:
"/home/user/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new
key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed
-- if you are prompted now it is to install the new keys
xetex@ubuntu's password:

Number of key(s) added: 1
```

Now try logging into the machine, with `ssh xetex@ubuntu` and check to make sure that only the key(s) you wanted were added.

Continued to page 44...